

Fig. 5: Flow-chart of the software program

the program is shown in Fig. 5. The trial version of Image Craft ICC AVR software is available on the Internet

at www.imagecraft.com/. Under 'device configuration' and press 'ok' button.

free of cost. It can be downloaded from the website 'www.imagecraft.com/'

Operation of ICC AVR IDE is simple. Create a folder, say, 'dcmotor.' Run the software and import the source code into the code editor screen by creating a new project file, say, 'EFY,' as follows: The 'EFY' file name will be displayed on the right-hand side of the screen. Add the source code in EFY. Open 'option' from 'project' menu. A 'compiler option' window opens. Here, select 'path' and provide proper path and location of the software.

Click 'compiler' menu and select the output format as 'COFF/HEX.' Next, select 'target' menu, choose 'Atmega8535' under 'device configuration' and press 'ok' button.

Now, from 'tools' menu, make the following settings in 'application builder' window:

Target CPU: M8535

Xtal: 4.00

Prescale: 16k

Compile the code by clicking 're-build' under 'project' menu. It will generate the hex file, listing file and other associated files. Burn the hex file into the microcontroller chip by using a programmer from Frontline Electronics. You can use any other equivalent Atmel programmer for the purpose.

Construction and testing

Assemble the circuit on a general-purpose PCB or on the PCB layout shown here. The actual-size, single-side PCB layout is shown in Fig. 6 and its component layout in Fig. 7.

After assembling the circuit on the PCB, check it for proper connections before switching on the power supply. Connect the outputs of the ULN2003 driver to the input terminals of each solidstate relay and the solidstate relays' outputs to the motor as shown in the schematic diagram.

You can use a power supply ranging from 24V to 200V DC depending on the DC motor rating used in your application. Make sure that the connecting wires to the

relays and the motor are capable of carrying high current during operation of the motor.

When power is switched on, LED1 should glow to indicate the presence of power in the circuit. Pressing switch S1 should display 'clockwise' message on the LCD and make the motor rotate in clockwise direction. Pressing switch S2 should display 'anticlockwise' on the LCD and make the motor rotate in an-

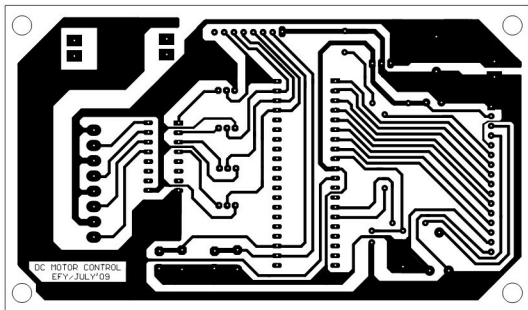


Fig. 6: Actual-size, single-side PCB for the AVR-based high-current DC motor controller